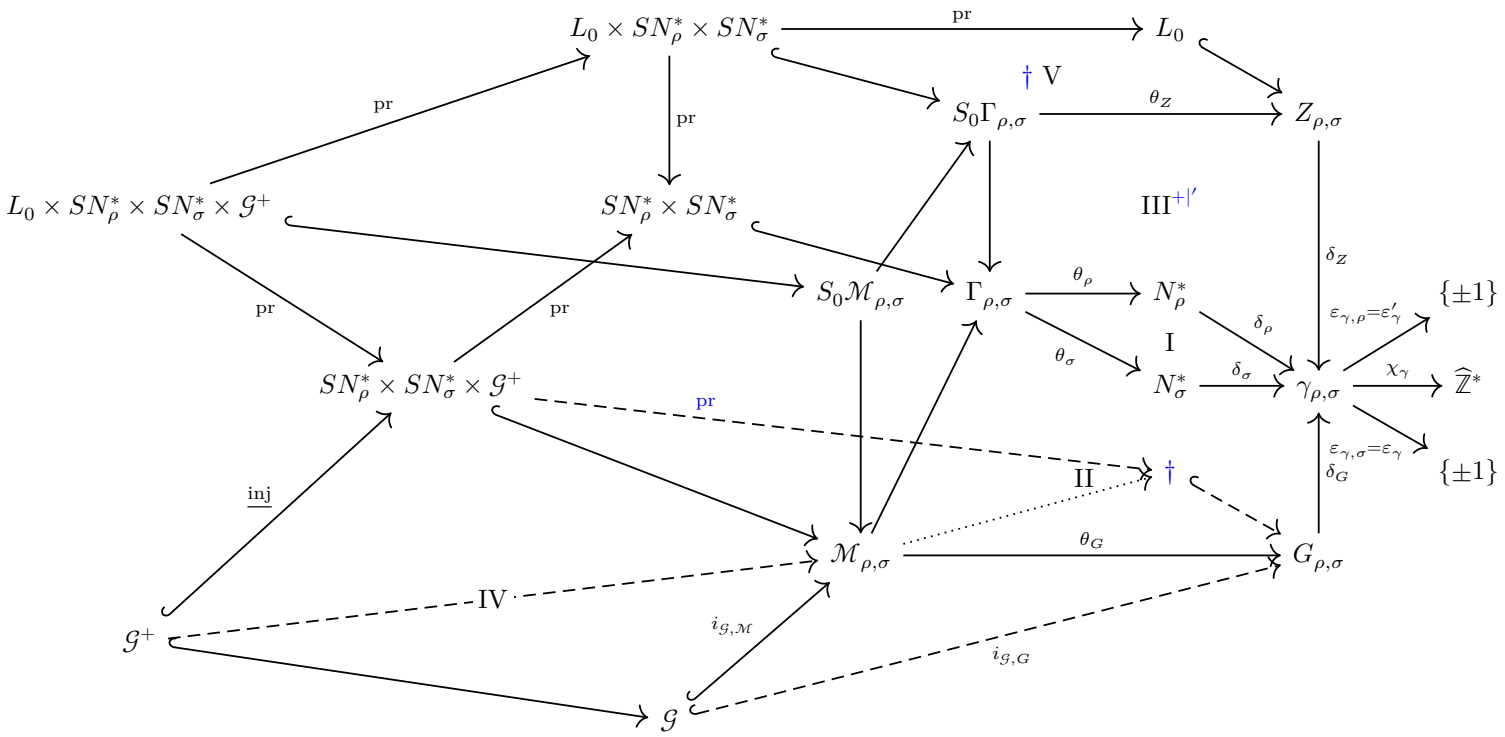
$$\begin{array}{lcccl}
\text{dimension 3} & \underline{\text{Cent}}(\mathcal{M}^{\circ}_{\rho,\sigma}) = R^{\circ}_{\rho,\sigma} & \xhookrightarrow[(\text{quotient } \mu_2\times\mu_2)]{} & R_{\rho,\sigma} & \text{dimension 3} \\
& \simeq ST_\rho \times ST_\sigma \times G_m & & \simeq S'T_\rho \times SN'_\sigma \times G_m & \\
& \uparrow & & \uparrow & \\
\text{dimension 2} & \underline{\text{Cent}}(\mathcal{M}_{\rho,\sigma}) \cap \mathcal{M}^{\circ}_{\rho,\sigma} & \xhookrightarrow[(\text{quotient } \mu_2)]{} & \underline{\text{Cent}}(\mathcal{M}_{\rho,\sigma}) & \text{dimension 2} \\
& = ST_\rho \times \mu_2 \times G_m & & \simeq S'T_\rho \times \dagger\mu_2 \times G_m) &
\end{array}$$

$$(8)\quad \mathcal{G}^{+\wedge} \xrightarrow{\text{sg|sgn}} \underset{\substack{\text{symmetric group}\\ \text{on three letters}}}{\mathcal{S}_3} = \mathcal{G}^{+\wedge}/\underset{\substack{\text{inverse}\\ \text{image}\\ \text{of } \Pi^{\wedge}_{0,3}\\ \text{in } \mathcal{G}^{+\wedge}}}{\widehat{\Pi}} \xrightarrow[\text{of a permutation}]{\text{sign}} \{\pm 1\}$$

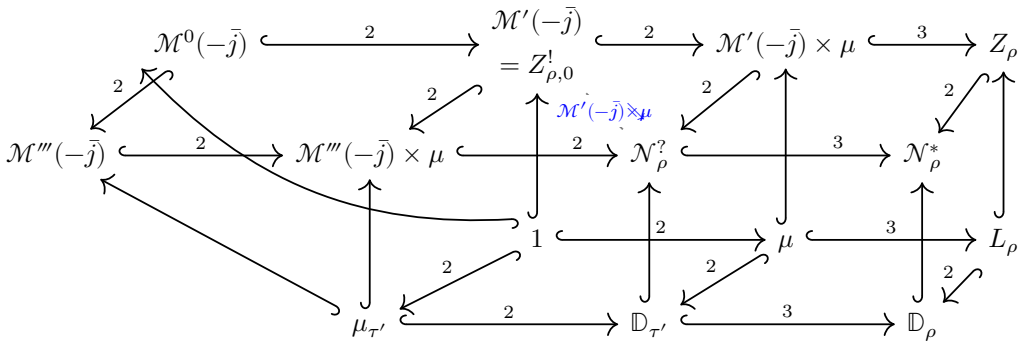
(62)

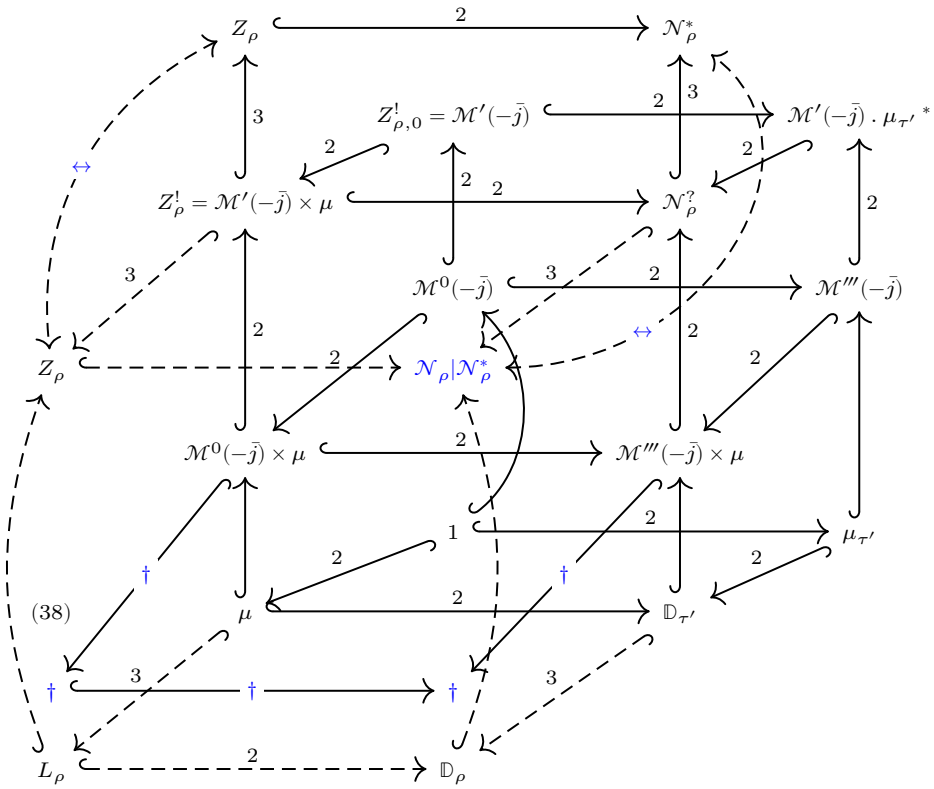


$\|$

$\{1, \tau'\} \cdot \mathcal{G}^+$

(37)


$\mathcal{M}^0(-\bar{j})$
$\mathcal{M}'(-\bar{j})$
$= Z^!_{\rho,0}$
$\mathcal{M}'(-\bar{j}) \times \mu$
Z_ρ
$\mathcal{M}'(-\bar{j})\times\mu$
$\mathcal{M}'''(-\bar{j})$
$\mathcal{M}'''(-\bar{j}) \times \mu$
$\mathcal{N}^?_\rho$
$\mathcal{N}^*_\rho$
1
μ
L_ρ
$\mu_{\tau'}$
$\mathbb{D}_{\tau'}$
$\mathbb{D}_\rho$
2
3


Z_ρ
$\mathcal{N}^*_\rho$
$Z^!_{\rho,0} = \mathcal{M}'(-\bar{j})$
$\mathcal{M}'(-\bar{j}) \cdot \mu_{\tau'}{}^*$
$Z^!_\rho = \mathcal{M}'(-\bar{j}) \times \mu$
$\mathcal{N}^?_\rho$
$\mathcal{M}^0(-\bar{j})$
$\mathcal{M}'''(-\bar{j})$
Z_ρ
$\mathcal{N}_\rho | \mathcal{N}^*_\rho$
$\mathcal{M}^0(-\bar{j}) \times \mu$
$\mathcal{M}'''(-\bar{j}) \times \mu$
$\mu_{\tau'}$
(38)
μ
$\mathbb{D}_{\tau'}$
L_ρ
$\mathbb{D}_\rho$